

Panel MS-AR(1): common intercept and linear trend

1985–2019, excluding the US, Germany, Japan, and China

1 Model

Seasonally adjusted real GDP per worker, 2026-09-17 vintage, 1985Q1 through 2019Q4. United States, Germany, and Japan are excluded. China is not in this vintage. No country random effects and no catch-up term.

$$y_{it} = a + g t + z_{it}, \tag{1}$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \tag{2}$$

$$s_{i,t+1} \sim \Pi(\cdot \mid s_{it}). \tag{3}$$

Three regimes. $E[z] = 0$ (the median regime mean is not pinned at 0). σ and ρ switch, with $|\rho| < 0.99$. Standard errors are delta-method from a numerical Hessian.

2 Common intercept and linear trend, 1985–2019, ex US, Germany, Japan, China

2026-09-17 SA panel, 1985Q1–2019Q4, excluding the US, Germany, and Japan. $y_{it} = a + gt + z_{it}$, $E[z] = 0$. 71 countries, 6378 observations, 1985-Q1–2019-Q4. Fitted countries: 71. Observations: 6378. Log-likelihood: 16903.85. Common intercept $a = -4.9602$ (0.0739). Common slope $g = 0.0061$ (0.0006).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-1.0912 (0.0891)	-0.8099 —	1.0343 (0.0721)
σ	0.0552 (0.0017)	0.0151 (0.0003)	0.0097 (0.0002)
ρ	0.9900 (0.0000)	0.9900 (0.0000)	0.9900 (0.0000)
$\Pi_0.$	0.8242 (0.0234)	0.1113 (0.0195)	0.0645 (0.0121)
$\Pi_1.$	0.0428 (0.0068)	0.9533 (0.0067)	0.0039 (0.0022)
$\Pi_2.$	0.0250 (0.0034)	0.0002	0.9748 (0.0036)
π	0.1581 (0.0161)	0.3786 (0.0355)	0.4633 (0.0376)

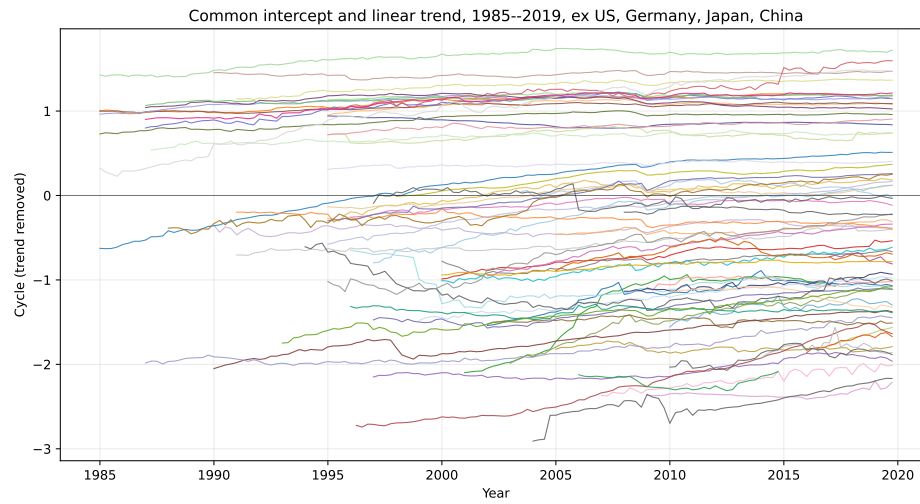


Figure 1: Country cycles after removing the common intercept and the linear trend.